OptiMOS™ IPOL TDA387xxx register map

Silicon revision 05/06(741)

About this document

Scope and purpose

The scope of this document applies to the register map information for single-output synchronous buck point-of-load (PoL) with PMBus, TDA38740/740A/725/725A silicon revision 05 and TDA38741 silicon revision 06. The corresponding IC_DEVICE_IDs are as follows:

TDA38740/740A, IC_DEVICE_ID – 0xA8, silicon revision 05
TDA38725/725A, IC_DEVICE_ID – 0xA9, silicon revision 05
TDA38741, IC_DEVICE_ID – 0xAA, silicon revision 06

The document contains applied conventions, register address locations, and function names and descriptions. This document provides internal register information to successfully configure the registers of the referenced device.

Intended audience

This document is intended for DC-DC power supply designers creating voltage regulator designs that require a PMBus communication PoL converter.

Note: The silicon revision can be read in register 0x00FC [15:8].

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1 16-bit register convention

The convention used throughout this document is:

- ‘h’ or ‘0x’ indicates that the number is expressed in hexadecimal
- ‘b’ indicates that the number is expressed in binary
- ‘d’ indicates that the number is expressed in decimal
- Bit 15 is the most significant bit (MSB) in the register
- Bit 0 is the least significant bit (LSB) in the register

Quite often, a 16-bit register is broken into several small pieces and each piece is used for a different function. The convention used in the document is in the following format:

Table 1 Example register

Address	Bits	Function names
0x0040	[15]	Reserved
	[14:8]	i2c_device_addr
	[7]	Reserved
	[6:0]	pmb_device_addr

The example in Table 1 indicates that register 0x0040 is broken into two functions:

- i2c_device_addr uses bits from 8 to 14
- pmb_device_addr uses bits from 0 to 6

While each register is 16 bits, they are addressed in an 8-bit format. For example, when referring to the register address 0x0020, it is referring to addresses 0x0020 and 0x0021, because each individual register is 8 bits wide.

From a user’s perspective, the TDA38740/740A/725/725A/741 register map is byte-addressable. The interface supports both 8- and 16-bit accesses, and registers can be written to or read from in either format.

Also, note the endianness of each data word. The format is using what is called the “Little-endian” convention, in which the higher byte is to the left and the lower byte is to the right. Therefore, when reading their value in the GUI for the base register 0x0000, which contains data 0xAA55, it should be interpreted as register 0x0001 contains 0xAA and register 0x0000 contains 0x55.

2 Register map sections

The register map is organized in two sections.

- Common registers, which contain config, trim, user, read-only (RO), read/write (R/W), and user read-only registers.
- PMBus registers.

Table 2 Register map sections

Section	Register	Base address
Common	Config	0x000
	Trim	0x004
	User	0x040
	Read-only	0x080
	Read/write	0x0c0
	User read-only	0x0f6
PMBus	PMBus	0x200

3 Common registers

3.1 OTP Config registers

Table 3 Config registers

ADDR offset	Contents	RST
0	[15:14] *Reserved [1:0]* [12:8] cnfg_first_user_image [5:0]: Specify the first user image that can be selected by the ADDR pin. This allows multi-config images to be reprogrammed. This register is zero-based. Set it to 0 at the first user image. rst:00000 [7:4] *Reserved [3:0]* [3:0] cnfg_num_user_configs [3:0]: Enables multiple user configuration images in non-volatile memory (NVM). This register defines the number of user images that can be selected by the ADDR pin at power-on. If the value specified by the ADDR pin exceeds this value, no user image will be restored at reset. This register is zero-based. Set it to 1 for two user images. Set it to 0 for one user image, which effectively disables the multi-config feature. rst:0000	0000
2	[15:8] cnfg_gpr_0 [7:0]: general-purpose register. rst:00000000 [7:0] *Reserved [7:0]*	0000

3.2 Trim registers

Table 4 Trim registers

Register	Address(h)	Bit field	Contents	Access	RST
Trim registers	04 to 3A	[15:0]	Reserved	R/W	

3.3 OTP user registers

Table 5 OTP user registers

Address offset	Contents	RST
40	[15] *Reserved* [14:8] i2c_device_addr [6:0]: Sets the I2C device address. In test mode, the device address is forced to 0x0A (7-bit address). If this register is set to 0, the test address (0x0A) will be selected. Reserved I2C addresses should be avoided: (0x00 to 0x07), 0x08, 0x0C, 0x28, 0x2F, 0x37, 0x61 (0x78 to 0x7F). All registers at this address are protected by the i2c_pmb_addr_lock register. rst:0010000 [7] *Reserved* [6:0] pmb_device_addr [6:0]: The chip PMBus address. Setting the address to zero disables the interface. Reserved addresses should be avoided, zone read, zone write. All registers at this address are protected by the i2c_pmb_addr_lock register. rst:1000000	1040
42	[15] Reserved [14] pmb_page_plus_latch_new_page: By default, the page specified in a PAGE_PLUS_READ or PAGE_PLUS_WRITE command is only used for that single transfer. The PAGE command value remains the same before and after a PAGE_PLUS command. This is the preferred behavior. When this bit is set, the PAGE command value will be changed to the page value specified in the PAGE_PLUS command. rst:0 [13] pmb_no_cml_on_bus_scan: Set this to prevent CML faults for too-short commands produced by unenlightened I2C/PMBus bus scans. rst:0 [12] pmb_nak_while_nvm_busy: When set, the PMBus interface will NAK all commands while the NVM is being accessed. rst:0 [11] pmb_salert_disable: If this bit is set, the SALERT signal will not be asserted. rst:0 [10] pmb_timeout_disable: For debug only. Disable the PMBus timeout monitor. rst:0 [9:8] pmb_pec_mode[1:0]: Set the PMBus PEC mode. In auto-PEC mode, PEC bytes are accepted and checked if they are received from the bus controller, and transmitted if requested by the bus controller. [7] i2c_input_dly_sel: Select a delay of either 360 ns (0) or 50 ns (1) on the delayed versions of the I2C SCL and SDA input signals. rst:0 [6] i2c_disable_addr_offset: When this bit is 0, the value in addr_pin_value is added to both i2c_device_addr and pmb_device_addr to set the device addresses. This allows multiple devices on the same bus to have unique device addresses. rst:0 [5] i2c_add_dout_hold: Set this bit to provide additional hold time (SCL to SDA) for outgoing I2C data. rst:0 [4] i2c_add_din_hold: Set this bit to provide additional hold time (SCL to SDA) for incoming I2C data. Enable this when the master provides insufficient hold time. Disable this when you want to run fast. rst:1 [3:2] i2c_scl_deglitch_en[1:0]: Enable the digital SCL deglitcher.	

<table>
<tr><th>Address offset</th><th>Contents</th><th>RST</th></tr>
<tr><td></td><td>
<table>
<tr><th>Value</th><th>Pulse limit</th></tr>
<tr><td>3</td><td>4 stage sync (20-30 ns)</td></tr>
<tr><td>2</td><td>3 stage sync (15-25 ns)</td></tr>
<tr><td>1</td><td>2 stage sync (10-20 ns)</td></tr>
<tr><td>0</td><td>no deglitch</td></tr>
</table>
rst:11
[1:0] i2c_sda_deglitch_en[1:0]: Enable the digital SDA deglitcher.
See i2c_scl_deglitch_en.
Note: both deglitchers should be set to the same value.
rst:11</td><td></td></tr>
<tr><td>48</td><td>[15:8] svid_vboot[7:0]: (741 only) Sets the Vboot during start-up with 20 mV resolution.</td><td></td></tr>
<tr><td>4A</td><td>[7:0] svid_vid_offset[7:0]: (741 only) Sets the Vboot during start-up with 1.25 mV resolution. Vboot voltage will be the sum of values set in svid_vboot and svid_vid_offset registers.</td><td></td></tr>
<tr><td>4B to 5A</td><td>Reserved</td><td></td></tr>
<tr><td>5C</td><td>[15:7] Reserved
[7] ov_asserts_lsfet: Select the OV fault source that engages the LSFET during an OV. Note that ov_neg_current_limit also affects the LSFET operation.
<table>
<tr><th>sel</th><th>description</th></tr>
<tr><td>1</td><td>Use the latched OV fault signal. The LSFET will be active until VOUT (via telemetry) drops below 250 mV.</td></tr>
<tr><td>0</td><td>Use the (qualified) OV fault signal from the analog comparator. The LSFET will be active until the OV fault signal deasserts.</td></tr>
</table>
rst:0
[6] ov_asserts_lsfet: Select whether to turn on the low-side FET in response to a high-side short. rst:1
[5:4] ov_neg_current_limit [1:0]: Set the limit for negative inductor current below, which turns off the LS FET during an overvoltage (OV) event.
<table>
<tr><td>ov_neg_current_limit</td><td>0</td><td>1</td><td>2</td><td>3</td></tr>
<tr><td>Current</td><td>No limit</td><td>-16 A</td><td>-20 A</td><td>-24 A</td></tr>
</table>
rst:00
[3:2] Reserved
[1] fccm_mode: Set this to force CCM mode. rst:0
[0] vid_dac_high_res: Enable high-resolution V_{OUT} DAC mode. The resolution of VOUT_COMMAND is set by the VOUT_MODE command, with a maximum resolution of 0.244 mV. The resolution of the V_{OUT} DAC is 1.25 mV. VOUT_COMMAND values are rounded to the nearest DAC value to determine the V_{OUT} set point. This register enables a higher DAC resolution (0.625 mV) by utilizing a DAC trim bit. rst:0</td><td>7040</td></tr>
<tr><td>5E</td><td>[15] relative_ovp_thresh_en: Use register relative_ovp_thresh to specify the overvoltage protection (OVP) threshold. This register overrides the PMBus commands. Set this bit to use register relative_ovp_thresh to specify the OV/UV threshold instead. This feature is available in all operating modes rst:0
[14:12] relative_ovp_thresh [2:0]: Specify the relative OVP threshold. It uses the same encoding as relative_uvp_thresh. rst:000</td><td>4400</td></tr>
</table>

Address offset	Contents	RST
	[11] relative_uvp_thresh_en: Use the register *relative_uvp_thresh* to specify the UVP threshold. This register overrides the PMBus commands. rst:0 [10:8] relative_uvp_thresh [2:0]: Specify the relative UVP threshold. The relative OVP and UVP thresholds are shown in the table.	

Relative OVP and UVP thresholds (ROVP/RUVP)

vout_scale_loop:	**1:1**			**1:2**		
relative_ovp thres_res:	**0 (50 mV step)**	**1 (100 mV step)**		**0 (100 mV step)**	**1 (200 mV step)**	
Setting	**ROVP/ RUVP (mV)**	**ROVP (mV)**	**RUVP (mV)**	**ROVP/ RUVP (mV)**	**ROVP (mV)**	**RUVP (mV)**
7	400	800	No UVP	800	1600	No UVP
6	350	700	No UVP	700	1400	No UVP
5	300	600	No UVP	600	1200	No UVP
4	250	500	No UVP	500	1000	No UVP
3	200	400		400	800	
2	150	300		300	600	
1	100	200		200	400	
0	50	100		100	200	

rst:000
[7] rovp_override_pin: Ignore the VBOOT pin, use the PMBus commands (or registers relative_[ovp,uvp]_thresh) to set the OVP and UVP thresholds. rst:0
[6] fovp_override_pin: Ignore the VBOOT pin, use register fixed_ovp_thresh. rst:0
[5] fccm_override_pin: Ignore the TON pin value, use register fccm_mode. rst:0
[4] ton_override_pin: Ignore the TON pin value, use PMBus command FREQUENCY_SWITCH (rounded to the eight supported frequencies) to set the switching frequency. rst:0
[3] vboot_override_pin: Ignore the VBOOT pin, use the VOUT_COMMAND value (PMBus mode) to specify the initial target V_{OUT} voltage. rst:0
[2] lcf_zero_override_pin: Ignore the ILIM pin value, use register loop_compensation_filter_zero_0 for the loop compensation zero value. rst:0
[1] ilim_aocp_override_pin: Ignore the ILIM pin value, use register aocp_thresh_sel for the analog OCP threshold. rst:0
[0] ilim_docp_override_pin: Ignore the ILIM pin value, use PMBus command IOUT_OC_FAULT_LIMIT for the digital OCP threshold. rst:0

60	[15] disable_relative_ovp: Ignore the relative OVP comparator. Always use the fixed OVP threshold. rst:0	1807

Address offset	Contents	RST

[14] disable_relative_uvp: Ignore the relative UVP comparator. rst:0
[13:12] blank_ov_sel [1:0]: Relative V_{OUT} OV faults can be blanked during voltage transitions.

Sel.	Description
3	Blank OV faults during DVID up or down (including V_{BOOT})
2	Blank OV faults during V_{BOOT} or DVID down
1	Blank OV faults during V_{BOOT}
0	No OV fault blanking

rst:01
[11:10] blank_uv_sel [1:0]: Relative V_{OUT} UV faults can be blanked during voltage transitions.

Sel.	Description
3	Blank UV faults during DVID up or down (including V_{BOOT})
2	Blank UV faults during DVID up (including V_{BOOT})
1	Blank UV faults during V_{BOOT}
0	No UV fault blanking

rst:10
[9:8] blank_oc_sel [1:0]: Digital I_{OUT} OC faults can be blanked during voltage transitions.

Sel.	Description
3	Blank OC faults during DVID up or down (including V_{BOOT})
2	Blank OC faults during DVID up (including V_{BOOT})
1	Blank OC faults during V_{BOOT}
0	No OC faults blanking

rst:00
[7] disable_digital_ocp: Disables the digital overcurrent protection (OCP). rst:0
[6:4] aocp_thresh_sel [2:0]: Analog OCP threshold select. On parts that support the ILIM pin, the OCP threshold is taken from the ILIM pin. On other parts, this register sets the OCP threshold.

Value	TDA38740A	TDA38725A
7	60 A	25 A
6	50 A	25 A
5	40 A	25 A
4	30 A	25 A
3	25 A	25 A
2	20 A	20 A
1	15 A	15 A
0	10 A	10 A

<table>
<tr><th>Address offset</th><th>Contents</th><th>RST</th></tr>
<tr><td></td><td>rst:000
[3] docp_from_aocp: Source the digital OC fault from the analog OC comparator. rst:0
[2:0] fixed_ovp_thresh [2:0]: (cot) Fixed (pre-) OVP threshold control bits.
<table>
<tr><th>Setting</th><th>vout_scale_loop 1:1</th><th>vout_scale_loop 1:2</th></tr>
<tr><td>7</td><td>2.85 V</td><td>5.7 V</td></tr>
<tr><td>6</td><td>2.2 V</td><td>4.4 V</td></tr>
<tr><td>5</td><td>1.8 V</td><td>3.6 V</td></tr>
<tr><td>4</td><td>1.5 V</td><td>3.0 V</td></tr>
<tr><td>3</td><td>1.35 V</td><td>2.7 V</td></tr>
<tr><td>2</td><td>1.2 V</td><td>2.4 V</td></tr>
<tr><td>1</td><td>1.0 V</td><td>2.0 V</td></tr>
<tr><td>0</td><td>0.8 V</td><td>1.6 V</td></tr>
</table>
rst:111</td><td></td></tr>
<tr><td>62</td><td>[15:0] Reserved [15:0]</td><td>7531</td></tr>
<tr><td>64</td><td>[15] Reserved
[14:12] loop_compensation_filter_zero_0 [2:0]: This register is used to position the internal loop compensation zero to aid with the crossover frequency and provide a phase boost. The register value changes from 0 to 7, which corresponds to an internal capacitance value ranging from 1 pF to 8 pF. rst:000
[11] d2a_enable_prebias: Enable pre-bias fix. rst:0
[10] Reserved
[9:8] gpr_usr_0[3:0]: General-purpose register. rst:0000
[7:4] afe_dig_offset_user [3:0]: (afed)
Range: -5 mV to 4.375 mV. Resolution: 0.625 mV.
rst:0000
[3] ldo_en_int_drv: Turns on the internal 5 V LDO from V_{IN}. rst:0
[2] input_power_vin_sel: Selects whether the input power calculation uses sensed V_{IN} (0) or fixed 12 V V_{IN} (1). rst:0
[1] iin_sense_mode_pmbus: By default, the PMBus READ_IIN command reports Malibu's local input current. When this bit is set, READ_IIN will report the input current selected by iin_sense_mode. rst:0
[0] iin_sense_pd: V_{IN} I_{sense}: power-down (disable). Assert for all applications not utilizing sensed input current (for power reduction, safe operating area (SOA) parking w/VCC pin). rst:0</td><td>0000</td></tr>
<tr><td>66</td><td>[15:0] Reserved</td><td>0001</td></tr>
<tr><td>68</td><td>[15:12] current_est_p_term [3:0]: P-term for current correction. [u-3.7] rst:0100
[11] gpr_usr_1: General-purpose register. rst:0
[10:0] current_est_dt_l [10:0]: Set this to the value of the inductor being used in the application. It is used to emulate the inductor current. The actual register value should be set to (10.4166666/inductor (in nH)) * 4096.
dt/L [u-1.12], dt = 10.4166 ns. rst:00011101101</td><td>40ED</td></tr>
</table>

<table>
<tr><th>Address offset</th><th>Contents</th><th>RST</th></tr>
<tr><td>6A</td><td>[15:12] current_est_i_term [3:0]: I-term for current correction. [u-15.19] rst:0001
[11:8] loadline_ignore_adc_window [3:0]: Ignore ADC output for (value * 2) samples after load step/release. rst:0100
[7] csa_sample_300mV_valley_in_dem: Forces the current sense amplifier (CSA) to sample 300 mV during diode emulation mode. rst:1
[6] loadline_range_sel: The loadline resistance is 0 mΩ to 10 mΩ by default. Set this register to extend the resistance by 5x. This effectively sets the resolution of VOUT_DROOP to either 5/256 or 25/256. rst:0
[5:4] loadline_current_sel [1:0]: Select the source of the output current for the loadline.
<table>
<tr><th>Sel.</th><th>Output current source</th></tr>
<tr><td>2, 3</td><td>Output current (via ADC)</td></tr>
<tr><td>1</td><td>Mean synthesized output current</td></tr>
<tr><td>0</td><td>Synthesized output current</td></tr>
</table>
rst:00
[3:0] loadline_bandwidth [3:0]: Loadline bandwidth, [u-4.9]. Controls the bandwidth of a filter that sits between the loadline voltage calculation and the DAC. In general, this value should be set as low as it can go without hurting the transient response. The filter bandwidth is (1 + reg_value) * 30 kHz. rst:0111</td><td>1487</td></tr>
<tr><td>6C</td><td>[15:12] loadline_follow_adc_window [3:0]: Follow ADC output for (value * 2) samples exiting ignore_adc state. rst:0100
[11] drv_hss_det_dis: (driver) Disables the HSS latch/event catcher and also the comparator output (if =0 then hss_raw and hss = 0). rst:0
[10] drv_uvlo_boot_dis: Disables the whole uvlo boot block (for faults and boot refresh). rst:0
[9] sst_pgood_off_on_fault: PMBus mode: deassert PGOOD immediately when a fault occurs. rst:0
[8] sst_immed_off_ramp: PMBus mode: when the output is turned off and ON_OFF_CONFIG[0] is set, by default VOUT will decay to 0 V. Set this bit to actively ramp VOUT to 0 V.
<table>
<tr><th>ON_OFF_CONFIG[0]</th><th>sst_immed_off_ramp</th><th>turn-off behavior</th></tr>
<tr><td>0</td><td>X</td><td>follow TOFF_DELAY and TOFF_FALL</td></tr>
<tr><td>1</td><td>0</td><td>decay to 0V</td></tr>
<tr><td>1</td><td>1</td><td>ramp to 0V immediately at TRANSITION_RATE slew rate</td></tr>
</table>
[7] Reserved
[6:0] low_power_when_enable_off [6:0]: Power down analog circuits when the output is not enabled, either by the EN pin or by PMBus command. Always set this register to 40h to disable the power-down feature to prevent any pre-bias start-up issues.</td><td>4068</td></tr>
</table>

<table>
<tr><th>Address offset</th><th>Contents</th><th>RST</th></tr>
<tr><td></td><td>
<table>
<tr><th>Bit</th><th>Circuit</th></tr>
<tr><td>[6]</td><td>Reserved</td></tr>
<tr><td>[5]</td><td>afea_en</td></tr>
<tr><td>[4]</td><td>ocp_zcd_en</td></tr>
<tr><td>[3]</td><td>drv_en</td></tr>
<tr><td>[2]</td><td>cotengine_core_en</td></tr>
<tr><td>[1]</td><td>csa_en: output current sense</td></tr>
<tr><td>[0]</td><td>adc_en: telemetry ADC</td></tr>
</table>
rst:1101000</td><td></td></tr>
<tr><td>6E</td><td>[15] uvp_rel_db_sel: Select either 80 ns (0) or 640 ns (1) debounce delay. rst:0
[14] ovp_rel_db_sel: Select either 80 ns (0) or 640 ns (1) debounce delay. rst:0
[13:12] ovp_fixed_db_sel [1:0]: Select a debounce duration for the analog fixed OVP signal.
<table>
<tr><th>Value</th><th>Debounce</th></tr>
<tr><td>3</td><td>1 µs</td></tr>
<tr><td>2</td><td>200 ns</td></tr>
<tr><td>1</td><td>100 ns</td></tr>
<tr><td>0</td><td>40 ns</td></tr>
</table>
rst:01
[11] cot_dem_fast_exit: The modulator will normally stay in diode emulation mode for a minimum of four switching cycles. This bit reduces the minimum number to 1. rst:0
[10:8] cot_mintoff_sel [2:0]: (cot) Programmable minimum off-time.
<table>
<tr><th>Sel</th><th>Min. off-time</th></tr>
<tr><td>7 to 5</td><td>150 ns</td></tr>
<tr><td>4</td><td>200 ns</td></tr>
<tr><td>3</td><td>270 ns</td></tr>
<tr><td>2</td><td>340 ns</td></tr>
<tr><td>1</td><td>400 ns</td></tr>
<tr><td>0</td><td>470 ns</td></tr>
</table>
rst:100
[7:4] Reserved
[3] vid_direct_clr_on_0v_vid: When set, vid_direct mode will be cleared when the target V_{OUT} is set to 0 V. rst:0
[2:0] period_range [2:0]: The loadline will assume a transient event if the measured switching frequency is outside the window (F_{sw} +/- transient_threshold). The loadline will ignore the ADC correction for loadline_ignore_adc_window cycles, then force ADC correction for loadline_follow_adc_window cycles. Q = 1/32 switching_period. Range = +/- [4, 11]/32.</td><td>1400</td></tr>
</table>

Address offset	Contents	RST
	<table><tr><th>Value</th><th>Transient threshold (+/- percent of switching period)</th></tr><tr><td>0</td><td>12.5%</td></tr><tr><td>1</td><td>15.6%</td></tr><tr><td>2</td><td>18.8%</td></tr><tr><td>3</td><td>21.9%</td></tr><tr><td>4</td><td>25.0%</td></tr><tr><td>5</td><td>28.1%</td></tr><tr><td>6</td><td>31.3%</td></tr><tr><td>7</td><td>34.4%</td></tr></table> rst:000	
70	[15:14] no_pulse_window[1:0]: If there are no PWM pulses for this many ADC samples (~1.1 MHz), then follow the isense ADC. rst:00 [13] gpio_ctrl_pin_db_sel: (741) Set the width of the debounce filter to either 120 ns (0) or 400 ns (1) for the ADDR/GPIO input pin. rst:0 [12] gpio_ctrl_en: (741) Enable VOUT control via the ADDR/GPIO pin. rst:0 [11] Reserved [10] drv_hss_db_sel: Set the width of the debounce filter to either 60 ns (0) or 220 ns (1) for the driver high-side short signal from the analog circuit. rst:0 [9] fixed_ovp_thresh_ext: Extends the maximum fixed OVP threshold to 6.0 V (normal mode is 5.7 V). rst:0 [8] relative_ovp_thresh_res: Select the resolution of the relative OV/UV comparator. 0 = 50 mV, 1 = 100 mV. rst:0 [7:6] *Reserved[1:0]* [5] read_protect_mode: Select the read protection mode for the config, trim, and user sections of the regmap. 0 = protection is enabled/disabled by the user password. 1 = protection is always enabled (the user password is ignored). rst:0 [4] write_protect_mode: Select the write protection mode for the user section of the regmap. *write_protect_mode* applies only to the user section (the trim section is write-protected by the trim password). See Access Protection for details. rst:0 [3:2] read_protect_section [1:0]: Select the regmap section to be read-protected. Reads from protected registers return 0xFFFF. *read_protect_section* applies to all sections (config, trim, and user). Note that only the user password is used for read protection, and it applies to config, trim, and user sections. The trim password is used for write protection only.rst:00 [1:0] write_protect_section [1:0]: Select the regmap section to be write-protected. Writes to protected registers will be silently ignored. *write_protect_section* applies to the user and config sections. The trim section is write-protected by trim_password. See Access Protection for details. rst:00	
72	[15:0] pmb_mfr_id [15:0]: MFR_ID (PMBus command). rst:0100100101000110	4946
74	[15:0] pmb_mfr_date [15:0]: MFR_DATE (PMBus command). rst:0	0000

Address offset	Contents	RST
76	[15:0] pmb_mfr_model [15:0]: MFR_MODEL [15:0] (PMBus command). If this register is zero, reading the command will return the value in *product_id_trim*. rst:0	0000
78	[15:0] pmb_mfr_revision [15:0]: MFR_REVISION [15:0] (PMBus command). If this register is zero, reading the command will return the silicon revision. rst:0	0000
7A	[15:0] user_password [15:0]: A 16-bit password that provides read/write protection for the user section in all regmaps. Password protection is enabled by the Protect Section and Protect Mode registers. See Access Protection for details. This register resets to zero, which is the default password. Once the password is set, access protection is enabled until *user_try_password* is set to the same value. rst:0	0000

3.4 Read_only registers

Address offset	Contents	
A8	[15:3] Infineon internal use only [2] nvm_user_loaded: Indicates that the USER registers have been restored from the NVM. [1] nvm_trim_loaded: Indicates that the TRIM registers have been restored from the NVM. [0] nvm_cnfg_loaded: Indicates that the CNFG registers have been restored from the NVM.	
AC	[9:4] nvm_user_image_sel[5:0]: Shows the user image selected by combining the MTP pin and register cnfg_num_user_configs. [3] Reserved [2:0] nvm_crc_err[2:0]: One-hot bitfield shows when a CRC error is detected when reading a CNFG (bit 0), TRIM (bit 1) and/or USER (bit 2) image from NVM.	
AE	[15:0] nvm_crc_value_msb[15:0]:	
B0	[15:0] nvm_crc_value_lsb[15:0]: Contains the CRC value from the most recently accessed image. After power-on in a fully programmed device, this will contain the CRC value from the USER image. After restoring a specific image (in any section) via the nvm_command register, this will contain the CRC value from the most recent image in that section. When an image is programmed, this will contain the calculated CRC value. If a restored image has a CRC error, this register will show the stored CRC value for the corrupt image.	
B2	[15:8] Infineon internal use only [7:3] nvm_ptr_cnfg[4:0]: A 5-bit bitfield indicating which CNFG images have been programmed. This register will be 0 for a new, unprogrammed part. It will be 0x1F when all 5 CNFG images are programmed. [2:0] nvm_ptr_trim[2:0]: A 3-bit bitfield indicating which TRIM images have been programmed. This register will be 0 for a new, unprogrammed part. It will be 0x7 when all 3 Trim images are programmed.	
B4	[15:0] nvm_ptr_user_xsb[15:0]: Bits [47:32] of the USER pointer	
B6	[15:0] nvm_ptr_user_msb[15:0]: Bits [31:16] of the USER pointer	

Address offset	Contents	
B8	[15:0] nvm_ptr_user_lsb[15:0]: Bits [15:0] of the USER pointer, a 48-bit bitfield that indicates which USER images have been programmed.	

3.5 Read_write registers

Address offset	Contents	
C2	[15:1] Infineon internal use only [0] soft_reset: Induce a soft reset. If the NVM contains a valid image, it is restored to the REGMAPs. This bit clears automatically.	0
D6	[15:0] nvm_command[15:0]: NVM Manager Command register.	0000

The NVM_COMMAND register

field	bits	encoding	description	
Done	15	1	Command has completed	
		0	Ready (or command is in progress). Set this bit to zero when issuing a new command.	
Error	14	1	An error occurred during a previous NVM operation. See the other NVM status registers for additional information.	
		0	The previous operation completed successfully	
Image	13:8		image number to be accessed (0x3F = use the most recent image)	
Section	7	rsv	This bitfield specifies the NVM section(s) to be accessed. Examples: to read only the TRIM section, set this field to 'b0010. To read the CNFG, TRIM and USER sections, set this field to 'b0111.	
	6	USER		
	5	TRIM		
	4	CNFG		
Opcode	3:0	15 - 9	*reserved*	
		8	TEST: write mode registers	the Trim password must be correct to execute test modes
		7	TEST: sense amp	
		6	TEST: cell stress	
		5	TEST: array clean	
		4	TEST: program word	
		3	TEST: read word	
		2	PROGRAM: write the values from the REGMAPs into the NVM	
		1	READ: restore the values from NVM to the REGMAPs	
		0	IDLE: nop	

DC	[15:0] fpga_rw_0[15:0]: Support register for FPGA emulation platform. Used as nvm_test_addr[10:0] during NVM Test Program and Test Read operations. Also used for programming.	
DE	[15:0] fpga_rw_1[15:0]: Support register for FPGA emulation platform. Used as nvm_test_addr[15:0] during NVM Test Program and Test Read operations. Also used for programming.	0000
E0	[15:0] fpga_rw_2[15:0]: Support register for FPGA emulation platform. Used as write/read data during NVM Test Program and Test Read operations. Also used for programming.	
E2	[15:0] fpga_rw_3[15:0]: Support register for FPGA emulation platform. Used as write/read data during NVM Test Program and Test Read operations. Also used for programming.	

3.6 User_read_only registers

Address offset	Contents
F6 to FA	Infineon internal use only
FC	[15:8] silicon_ver [7:0]: Hardwired silicon revision
	[7:0] product_id [7:0]: The product ID (same as the value stored in the NVM)
FE	[15:8] i2c_window [7:0]: The 8 MSBs of the I2C register address. See the communications interface description for more information.

4 PMBus register map section

Table 6 Legend

Legend	
Normal	The command is supported, and its value is stored in a register at the specified offset. To address a register, add the "ADDR offset" (first column) to the base address for the loop.
rsv/unsup	The command is either not supported, or is a reserved PMBus command
no_reg	The command is supported but has no register storage. If you read the value at the corresponding address (via the I2C bus, or the D0 PMBus commands), the data is undefined.

Base address – 0x0200

- The "ADDR offset" in the Table 7 (shown in 3-bit hex code) represents the offset that needs to be added to the base address of the particular loop to access the registers associated with that loop.

Table 7 OTP PMBus registers

CMD	ADDR offset	Bits	Contents	Res.	Reset value
00	0x000		**Page:** there is no regmap storage for this command.	0x00, 0xFF	0x00
01	0x002	15:8	*Reserved*		
		7:0	**Operation:** Malibu supports all legal values for the OPERATION command, in accordance with the CAPABILITIES command		0x0

Bits	Description	Value	Operation
7:6	On/off control	1x	On
		01	Turn off with TOFF_DELAY and TOFF_FALL
		00	Turn off immediately *(overrides ON_OFF_CONFIG [0])*
5:4	Voltage command source	11	AVS_VOUT_COMMAND – not supported in Malibu
		10	VOUT_MARGIN_HIGH
		01	VOUT_MARGIN_LOW
		00	VOUT_COMMAND
3:2	Margin fault response	10	Ignore faults caused by VOUT_MARGIN_HIGH or VOUT_MARGIN_LOW
		01	Act on faults caused by VOUT_MARGIN_HIGH or VOUT_MARGIN_LOW

CMD	ADDR offset	Bits	Contents				Res.	Reset value
					11,00	Normal operation		
			1	AVS transition control	Not used in Malibu			
			0	*reserved*				
02	0x004	15:8	*Reserved*					
		7:0	**on_off_config:** The current value of the ON_OFF_CONFIG command.					0x17
			7:5	*Reserved*	000			
			4	Operation control	1	Power-up controlled by EN pin and/or OPERATION command		
					0	Power-up when power is present (ignore EN and OPERATION)		
			3	Response to OPERATION command	1	OPERATION command is required to power up		
					0	OPERATION command is ignored		
			2	Response to EN pin	1	EN pin is required to power up		
					0	EN pin is ignored		
			1	EN pin polarity	1	Active high		
					0	Active low		
			0	EN pin turn-off behavior	1	Turn off immediately		
					0	Turn off with TOFF_DELAY and TOFF_FALL		
03	0x006		**Clear_faults:** The CLEAR_FAULTS command is write-only. It clears SALERT and all status registers in the current page. There is no register storage for this command.					
04	0x008		**Phase:** command not supported					
05	0x00a		**page_plus_write:** Perform a write on the PMBus page specified in this command. The page will revert to its previous value unless the *regmap_common.pmb_page_plus_latch_new_page* is asserted.					
06	0x00c		**page_plus_read:** Perform a read on the PMBus page specified in this command. The page will revert to its previous value unless the *regmap_common.pmb_page_plus_latch_new_page* is asserted.					
07	0x00e		Reserved PMBus command					
08	0x010		Reserved PMBus command					
09	0x012		Reserved PMBus command					

CMD	ADDR offset	Bits	Contents	Res.	Reset value
0A	0x014		Reserved PMBus command		
0B	0x016		Reserved PMBus command		
0C	0x018		Reserved PMBus command		
0D	0x01a		Reserved PMBus command		
0E	0x01c		Reserved PMBus command		
0F	0x01e		Reserved PMBus command		
10	0x020	15:8	*Reserved*		
		7:0	**write_protect:** *Set the PMBus write protection mode.* Value: Write-protected commands 0x80: All except WRITE_PROTECT 0x40: All except WRITE_PROTECT, OPERATION, PAGE 0x20: All except WRITE_PROTECT, OPERATION, PAGE, ON_OFF_CONFIG, VOUT_COMMAND 0x03: All writes are disabled 0x02: All except VOUT_COMMAND 0x00: All writes are enabled	0x80, 0x40, 0x20, 0x03, 0x02, 0x00	0x0
11	0x022		**store_default_all:** There is no regmap storage for this command		
12	0x024		**restore_default_all:** Write-only command that entices TDA38740/740A/725/725A/741 to restore the contents of the most recent user image from NVM into the registers. Note that the function of this command is identical to RESTORE_USER_ALL.		
13	0x026		**store_default_code:** Command not supported		
14	0x028		**restore_default_code:** Command not supported		
15	0x02a		**store_user_all:** Write-only command that programs the contents of the NVM-backed USER registers into the NVM array. When the programming is complete, nvm_command[15] will be set to 1 by the hardware.		
16	0x02c		**restore_user_all:** Write-only command that causes TDA38740/740A/725/725A/741 to restore the contents of the most recent user image from NVM into the registers. Note that the function of this command is identical to RESTORE_DEFAULT_ALL.		
17	0x02e		**store_user_code:** Command not supported		
18	0x030		**restore_user_code:** Command not supported		
19	0x032		**capability:** There is no register storage for this read-only command. The value is provided by the PMBus interface logic.		
1A	0x034		**query:** Command not supported		

CMD	ADDR offset	Bits	Contents	Res.	Reset value
1B	0x036		**smbalert_mask:** Provides indirect access to the STATUS_MASK registers. There is no register storage for this command. The status masks are stored in the MSB of the status registers.		
1C	0x038		Reserved PMBus command		
1D	0x03a		Reserved PMBus command		
1E	0x03c		Reserved PMBus command		
1F	0x03e		Reserved PMBus command		
20	0x040	15:8	*Reserved*		
		7:0	**vout_mode:** The exponent of the V_{OUT}-related commands (for example, VOUT_COMMAND, VOUT_MARGIN_HIGH, VOUT_MARGIN_LOW, VOUT_MAX, VOUT_OV_FAULT_LIMIT, VOUT_UV_FAULT_LIMIT, POWER_GOOD_ON, POWER_GOOD_OFF and READ_VOUT). TDA38740/740A/725/725A/741 supports exponents of -8, -9, -10, -11 and -12. Command format (binary): mmm_eeeee mmm = mode of the V_{OUT}-related commands (Malibu only supports 000: unsigned linear-16) eeeee = 2's complement exponent for the 16-bit mantissas specified by the V_{OUT}-related commands Example: vout_mode = 0x14 sets the exponent to -12 for all V_{OUT}-related commands	0x14, 0x15, 0x16, 0x17, 0x18	0x18
21	0x042	15:0	**vout_command:** Sets the output voltage in volts (not VID codes). The resolution for all V_{OUT}-related commands is set by VOUT_MODE.	vout_mode	0x010D
22	0x044		**vout_trim:** Command not supported		
23	0x046		**vout_cal_offset:** Command not supported		
24	0x048	15:0	**vout_max:** Sets an upper limit on the commanded output voltage. Setting the output voltage to a value higher than the value in this command will cause a VOUT_MAX warning.	vout_mode	0x0500
25	0x04a	15:0	**vout_margin_high:** Specifies the target output voltage when the OPERATION command is set to *margin_high*.	vout_mode	0x0135
26	0x04c	15:0	**vout_margin_low:** Specifies the target output voltage when the OPERATION command is set to *margin_low*.	vout_mode	0x0100
27	0x04e	15:0	**vout_transition_rate:** This command specifies the output slew rate during regulation. It does not apply to the initial voltage ramp from 0 V (turning on, controlled by *TON_RISE*) or the ramp to 0 V (turning off, controlled by *TOFF_FALL*).	u7.3	0xE808
28	0x050	15:0	**vout_droop:** Specify the loadline resistance value. The resolution is set by loadline_range_sel to either 5/256 mΩ or 25/256 mΩ. With *loadline_range_sel*=0, a value of 0x33 (51 decimal) sets the loadline to 0.996 mΩ.		0x0

CMD	ADDR offset	Bits	Contents	Res.	Reset value
			With *loadline_range_sel*=1, a value of 0x0A (10 decimal) sets the loadline to 0.9765 mΩ.		
29	0x052	15:0	**vout_scale_loop:** TDA38740/740A/725/725A/741 supports 1:1 (0xE808) and 1:2 (0xE804) external resistor-divider ratios.	0xE808, 0xE804	0xE808
2A	0x054		**vout_scale_monitor:** Command not supported		
2B	0x056		**vout_min:** Sets a lower limit on the commanded output voltage. Setting the output voltage to a value lower than the value in this command will cause a VOUT_MAX_MIN warning.	vout_mode	0x0000
2C	0x058		Reserved PMBus command		
2D	0x05a		Reserved PMBus command		
2E	0x05c		Reserved PMBus command		
2F	0x05e		Reserved PMBus command		
30	0x060		**coefficients:** Command not supported		
31	0x062		**pout_max:** Command not supported		
32	0x064		**max_duty:** Command not supported		
33	0x066	15:0	**frequency_switch:** Sets the PWM switching frequency. Range is 400 kHz to 2000 kHz with 200 kHz steps with the exception of 1600 kHz.	u10.0 or u11.-1	0x0320
34	0x068		**Power_mode:** Command not supported		
35	0x06a	15:0	**vin_on:** Sets the V_{IN} voltage above which the output will be allowed to operate. VIN_ON and VIN_OFF define how the *vin_uv_fault* sticky bit is set. There is no corresponding bit in the STATUS registers. To set the *vin_uv_fault* sticky bit, V_{IN} must exceed VIN_ON and then fall below V_{IN_OFF}.	u5.1	0xF810
36	0x06c	15:0	**vin_off:** Sets the V_{IN} voltage below which the output will refuse to operate.	u5.1	0xF80E
37	0x06e		**interleave:** Command not supported		
38	0x070	15:0	**iout_cal_gain:** Set current sense gain. 2's complement.	s-1.7	0xC800
39	0x072	15:0	**iout_cal_offset:** Set the current sense offset. 2's complement.	s7.4	0xE000
3A	0x074		**fan_config_1_2:** Command not supported		
3B	0x076		**fan_command_1:** Command not supported		
3C	0x078		**fan_command_2:** Command not supported		
3D	0x07a		**fan_config_3_4:** Command not supported		
3E	0x07c		**fan_command_3:** Command not supported		
3F	0x07e		**fan_command_4:** Command not supported		
40	0x080	15:0	**vout_ov_fault_limit:** This command specifies an absolute (that is, non-relative) threshold for OVP. The OV threshold is the difference	vout_mode	0x0133

CMD	ADDR offset	Bits	Contents	Res.	Reset value
			between this value and the target V_{OUT}. The resolution of this command is specified by VOUT_MODE.		
41	0x082	15:8	*Reserved*		
		7:0	**vout_ov_fault_response:** Specifies our response to a V_{OUT} OV fault. Command format (binary): xx_rrr_ddd xx = response (00 = ignore, 10 = shutdown/retry, 11 = restart when fault condition clears (restart not supported in Part)) rrr = number of retries (0 to 6, 7 = retry forever) ddd = delay between retries (1 ms to 8 ms)	0x00, 0x3F, 0x80-0xBF	0x80
42	0x084	15:0	vout_ov_warn_limit: Command not supported		
43	0x086	15:0	vout_uv_warn_limit: Command not supported		
44	0x088	15:0	**vout_uv_fault_limit:** Similar to VOUT_OV_FAULT_LIMIT. Resolution is specified by VOUT_MODE.	vout_mode	0x0000
45	0x08a	15:8	*Reserved*		
		7:0	**vout_uv_fault_response:** Specifies our response to a V_{OUT} UV fault. Command format (binary): xx_rrr_ddd xx = response (00 = ignore, 10 = shutdown/retry, 11 = restart when fault condition clears (restart not supported in Part)) rrr = number of retries (0 to 6, 7 = retry forever) ddd = delay between retries (1 ms to 8 ms)	0x00, 0x3F, 0x80-0xBF	0x00
46	0x08c	15:0	**iout_oc_fault_limit:** Sets the I_{OUT} OC fault threshold.	u6.4	0xE140
47	0x08e	15:8	*Reserved*		
		7:0	**iout_oc_fault_response:** Specifies TDA38740/740A/725/725A/741's response to an I_{OUT} OC fault. Command format (binary): xx_rrr_ddd xx = response ('b11 = shut down and respond per [5:3] and [2:0]) rrr = number of retries (0 to 6, 7 = retry forever) ddd = delay between retries (1 ms to 8 ms) Supported: 'b11_rrr_ddd (shut down, respond per [5:3])	0xC0-0xFF	0xF0
48	0x090		**iout_oc_lv_fault_limit:** Command not supported		
49	0x092		**iout_oc_lv_fault_response:** Command not supported		
4A	0x094		**iout_oc_warn_limit:** Command not supported		
4B	0x096		**iout_uc_fault_limit:** Command not supported		
4C	0x098		**iout_uc_fault_response:** Command not supported		
4D	0x09a		Reserved PMBus command		
4E	0x09c		Reserved PMBus command		
4F	0x09e	15:0	**ot_fault_limit:** Sets the overtemperature fault threshold.	u8.0	0x008C

CMD	ADDR offset	Bits	Contents	Res.	Reset value
50	0x0a0	15:8	*Reserved*		
		7:0	**ot_fault_response:** Specifies our response to an overtemperature fault. Command format (binary): xx_rrr_ddd xx = response (00 = ignore, 10 = shut down/retry, 11 = restart when fault condition clears (restart not supported in Malibu)) rrr = number of retries (0 to 6, 7 = retry forever) ddd = delay between retries (1 ms to 8 ms)	0x00, 0x3F, 0x80-0xBF	0xB8
51	0x0a2	15:0	**ot_warn_limit:** Sets the overtemperature warning threshold	u8.0	0x0078
52	0x0a4		**ut_warn_limit:** Command not supported		
53	0x0a6		**ut_fault_limit:** Command not supported		
54	0x0a8		**ut_fault_response:** Command not supported		
55	0x0aa	15:0	**vin_ov_fault_limit:** Sets the V_{IN} overvoltage fault threshold.	u6.4	0xE0F0
56	0x0ac	15:8	*Reserved*		
		7:0	**vin_ov_fault_response:** Specifies our response to an input OV temperature fault. Command format (binary): xx_rrr_ddd xx = response (00 = ignore, 10 = shutdown/retry, 11 = restart when fault condition clears (restart not supported in Malibu)) rrr = number of retries (0 to 6, 7 = retry forever) ddd = delay between retries (1 ms to 8 ms) specifies TDA38740/740A/725/725A/741's behavior in response to a V_{IN} OV fault. TDA38740/740A/725/725A/741 supports 0x00 (ignore) and 0x80 (shutdown) responses.	0x00, 0x3F, 0x80-0xBF	0x00
57	0x0ae		**vin_ov_warn_limit:** Command not supported		
58	0x0b0	15:0	**vin_uv_warn_limit:** Sets the V_{IN} UV warning threshold	u6.4	0xE000
59	0x0b2		**vin_uv_fault_limit:** Command not supported		
5A	0x0b4		**vin_uv_fault_response:** Command not supported		
5B	0x0b6		**iin_oc_fault_limit:** Command not supported		
5C	0x0b8		**iin_oc_fault_response:** Command not supported		
5D	0x0ba	15:0	**iin_oc_warn_limit:** Command not supported		

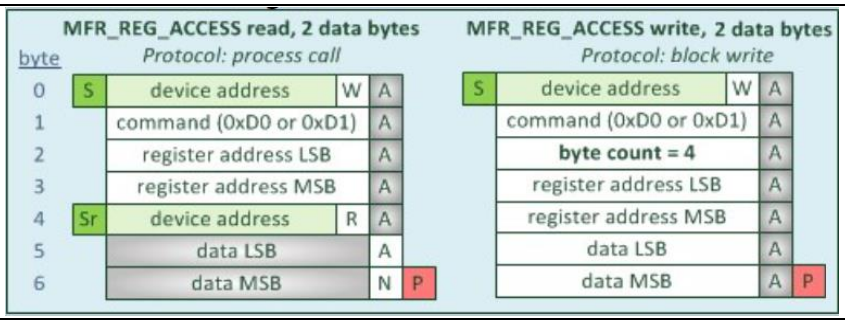
CMD	ADDR offset	Bits	Contents	Res.	Reset value
5E	0x0bc	15:0	**power_good_on:** Sets the output voltage above which the PowerGood indication is asserted. Resolution is specified by VOUT_MODE.	vout_mode	0x0100
5F	0x0be	15:0	**power_good_off:** Sets the output voltage below which the PowerGood indication is deasserted. Resolution is specified by VOUT_MODE.	vout_mode	0x00E8
60	0x0c0	15:0	**ton_delay:** Specifies the time (in ms) from when the output is enabled until the voltage starts to ramp.	u7.1	0xF800
61	0x0c2	15:0	**ton_rise:** Specifies the time (in ms) from when the output is enabled until it reaches the target voltage. This value is only used for the initial ramp. *VOUT_TRANSITION_RATE* controls the slew rate during regulation.	u7.1	0xF802
62	0x0c4	15:0	**ton_max_fault_limit:** Command not supported		
63	0x0c6	15:8	*Reserved*		
		7:0	**ton_max_fault_response:** Command not supported		
64	0x0c8	15:0	**toff_delay:** Specifies the time (in ms) from when the output is disabled until the voltage begins to ramp to 0 V.	u7.1	0xF800
65	0x0ca	15:0	**toff_fall:** Specifies the time (in ms) from when the output is disabled to when the voltage is commanded to 0 V. This value is only used when turning the output off. *VOUT_TRANSITION_RATE* controls the slew rate during regulation.	u7.1	0xF002
66	0x0cc		**toff_max_warn_limit:** Command not supported		
67	0x0ce		Reserved PMBus command		
68	0x0d0		**pout_op_fault_limit:** Command not supported		
69	0x0d2		**pout_op_fault_response:** Command not supported		
6A	0x0d4		**pout_op_warn_limit:** Command not supported		
6B	0x0d6		**pin_op_warn_limit:** Command not supported		
6C	0x0d8		Reserved PMBus command		
6D	0x0da		Reserved PMBus command		
6E	0x0dc		Reserved PMBus command		
6F	0x0de		Reserved PMBus command		
70	0x0e0		Reserved PMBus command		

CMD	ADDR offset	Bits	Contents	Res.	Reset value
71	0x0e2		Reserved PMBus command		
72	0x0e4		Reserved PMBus command		
73	0x0e6		Reserved PMBus command		
74	0x0e8		Reserved PMBus command		
75	0x0ea		Reserved PMBus command		
76	0x0ec		Reserved PMBus command		
77	0x0ee		Reserved PMBus command		
78	0x0f0		**status_byte:** The eight LSBs of STATUS_WORD. There is no regmap storage for this command.		
79	0x0f2	15:0	**status_word:** 16-bit status summary Bit: STATUS_WORD 7: V_{OUT} 6: I_{OUT} 5: INPUT 4: MFR_SPECIFIC 3: POWER_GOOD# 2: *Unused* 1: *Unused* 0: *Unused* Bit: STATUS_BYTE 7: *Unused* 6: OFF 5: VOUT_OV_FAULT 4: IOUT_OC_FAULT 3: *Unused* 2: TEMPERATURE 1: CML 0: NONE_OF_THE_ABOVE		0x0
7A	0x0f4	15:8	**status_vout_mask:** Status mask bits. Typically accessed indirectly via the SMBALERT_MASK command.		0x0
		7:0	**status_vout:** Fault and warning information related to V_{OUT} Bit: Status 7: VOUT_OV_FAULT 6: *Unused* 5: *Unused*		0x0

CMD	ADDR offset	Bits	Contents	Res.	Reset value
			4 VOUT_UV_FAULT 3 VOUT_MAX_Warning 2 *Unused* 1 *Unused* 0 *Unused*		
7B	0x0f6	15:8	**status_iout_mask:** Status mask bits. Typically accessed indirectly via the SMBALERT_MASK command.		0x0
		7:0	**status_iout:** Fault and warning information related to I_{OUT} Bit: Status 7: IOUT_OC_FAULT 6: *Unused* 5: *Unused* 4:0: *Unused*		0x0
7C	0x0f8	15:8	**status_input_mask:** Status mask bits. Typically accessed indirectly via the SMBALERT_MASK command.		0x0
		7:0	**status_input:** Fault and warning information related to V_{IN} Bit: Status 7: VIN_OV_FAULT 6: *Unused* 5: *Unused* 4: *Unused* 3: Off due to VIN_UV 2:0: *Unused*		0x0
7D	0x0fa	15:8	**status_temperature_mask:** Status mask bits. Typically accessed indirectly via the SMBALERT_MASK command.		0x0
		7:0	**status_temperature:** Fault and warning information related to temperature Bit: Status 7: OT_FAULT 6: OT_WARNING 5:0: *Unused*		0x0
7E	0x0fc	15:8	**status_cml_mask:** Status mask bits. Typically accessed indirectly via the SMBALERT_MASK command.		0x0

CMD	ADDR offset	Bits	Contents	Res.	Reset value
		7:0	**status_cml:** Fault and warning information related to PMBus communications. Bit: Status 7: Invalid/Unsupported command 6: Invalid/Unsupported data, or *send_too_many_bytes* 5: PEC check failed 4: OTP store/restore error 3: *Unused: memory fault* 2: *Unused: reserved* 1: *read_bit_set_in_addr* OR *send_too_few_bits* OR *read_too_few_bits* OR *command_too_short* OR *read_too_many_bytes* 0: *Unused: other memory/logic fault*		0x0
7F	0x0fe		**status_other:** Command not supported		
80	0x100		**status_mfr_specific_mask:** Command not supported		
			status_mfr_specific: Command not supported		
81	0x102		**status_fans_1_2:** Command not supported		
82	0x104		**status_fans_3_4:** Command not supported		
83	0x106		Reserved PMBus command		
84	0x108		Reserved PMBus command		
85	0x10a		Reserved PMBus command		
86	0x10c		**read_ein:** Command not supported		
87	0x10e		**read_eout:** Command not supported		
88	0x110	15:0	**read_vin:** Telemetry: input voltage. Resolution: 1/32 V	u5.5	0x0
89	0x112	15:0	**read_iin:** Telemetry: input current. Resolution: auto-adjust from 1 A to 1/32 A	u10.0 to u5.5	0x0
8A	0x114		**read_vcap:** Command not supported		
8B	0x116	15:0	**read_vout:** Telemetry: output voltage. Resolution is set by VOUT_MODE.	vout_mode	0x0
8C	0x118	15:0	**read_iout:** Telemetry: output current. Resolution: 1/16 A.	u6.4	0x0
8D	0x11a	15:0	**read_temperature_1:** Telemetry: temperature. Resolution: 1°C.	u9.0	0x0
8E to 95	0x11c to 0x12a		**read_temperature_2:** Command not supported		

CMD	ADDR offset	Bits	Contents	Res.	Reset value
96	0x12c	15:0	**read_pout:** Telemetry: output power. Resolution: 1 W.	u8.0	0x0
97	0x12e	15:0	**read_pin:** Telemetry: input power. Resolution: auto-adjust from 1 W to 1/8 W.	u10.0 to u8.2	0x0
98	0x130		**pmbus_revision:** There is no regmap storage for this command		
99	0x132		**mfr_id:** Manufacturer ID, 2 bytes, R/W. The command value is common to all pages and resides in regmap_common. Default = 0x4946.		
9A	0x134		**mfr_model:** Manufacturer model, 2 bytes, R/W. The command value is common to all pages and resides in regmap_common. Returns the value in regmap_common.pmb_mfr_model at reset, and if written to 0x0.		
9B	0x136		**mfr_revision:** Manufacturer revision, 2 bytes, R/W. The command value is common to all pages and resides in regmap_common. Returns the value in regmap_common.pmb_mfr_revision at reset, and if written to 0x0.		
9C to AC	0x138 to 0x158		Command not supported		
AD	0x15a		**ic_device_id:** Device ID. This command maps to regmap_common. product_id.		
AE	0x15c		**ic_device_rev:** Device revision. This command maps to the hardwired silicon revision number, also found in regmap_common. silicon_ver.		
AF to C0	0x15e to 0x180		Command not supported		
C1	0x182		**mfr_vendor_info_1:** Vendor info, 2 bytes, read-only. Returns the product_id and silicon revision.		0x8200
C2	0x184		**mfr_vendor_info_2:** Vendor info, 2 bytes, R/W. Available for vendor use. Default= 0x01.		0x0001
C3 to CF	0x186 to 0x19E		Command not supported		
D0	0x1a0		**mfr_reg_access:** This command provides read/write access to non-PMBus registers. This command uses the SMBus Process Call protocol for reads, and the Block Write protocol for writes, as shown below. The first address byte contains the address offset for the target register. The second address byte contains the base address of the target regmap.		

CMD	ADDR offset	Bits	Contents	Res.	Reset value
					
D1 to FF	0x1a6 to 0x1fe		Command not supported		

5 Formats for reading and writing

TDA38740/740A/725/725A/741 supports the I2C and PMBus bus protocols. Both protocols use the same physical interface (the SCL and SDA pins). The device address provided in the transfer's first serial byte selects the active interface. The registers in REGMAP Common have traditionally been accessed using the I2C protocol, and the PMBus Command registers have traditionally been accessed using the PMBus protocol. Note that the Part allows access to all registers using one of the protocols (with two exceptions).

TDA38740/740A/725/725A/741 supports I2C packets that include 8-register address bits (not the full 16-address bits that are used internally). The bus interface logic provides a "window" register to store 8-additional address bits. Thus, the entire 64 KB address range can be addressed, including the registers that hold the PMBus command values. The user first writes the 8-high-order address bits to the window register and then accesses the registers within that window. The window register resides at address 0xFF within each window and is only accessible via the I2C interface.

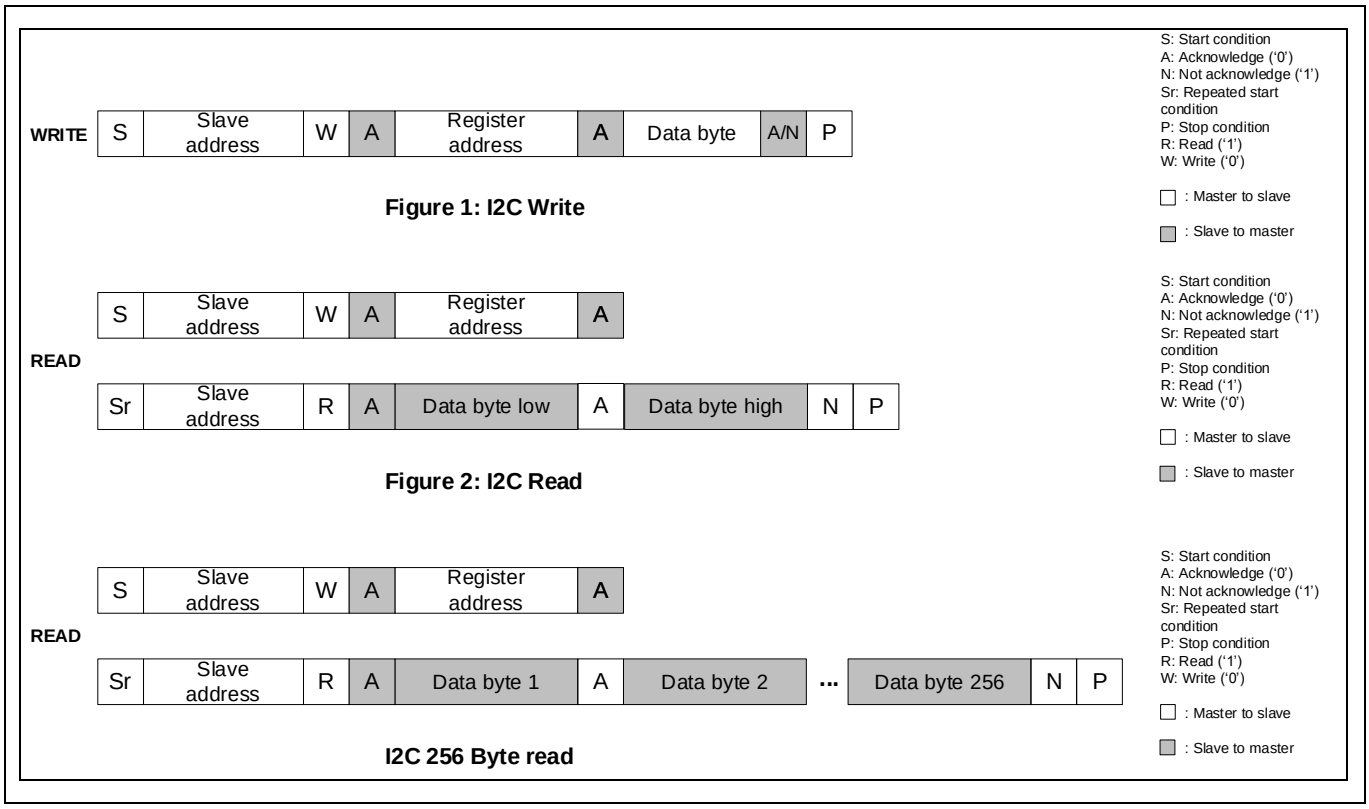


Figure 1 Format for reading and writing

Note: Data1, data2, etc. are the data bytes read from the device, with data1 corresponding to the data returned from address 0x0000, data2 corresponding to the data returned from address 0x0001, etc.

6 Register access protection

TDA38740/740A/725/725A/741 provides multiple register read and write access protection mechanisms. TDA38740/740A/725/725A/741 register maps are divided into multiple “sections” (config, trim, and user). Separate access protection is provided for each register map section. Protection is also provided for specific registers within a section, such as the I2C device address register.

- Passwords protect a regmap section.
- Lock bits protect specific registers, independent of the regmap sections.

In some cases, the protection mechanisms overlap. All protection mechanisms must be disabled to access a protected register.

6.1 Lock bits

There are three lock bits that provide write protection to specific registers. They are independent of the mode, section, and password mechanisms (see Table 8). The lock bits provide write protection only; reads are unaffected. The locks are enabled and disabled by simply writing to the appropriate bit in the register map.

Table 8 Lock bit protection

Lock bit protection			
		Protected address offsets *are 16-bit addresses ranges are inclusive*	
Lock bit	**Description**	**Common**	**PMBus**
trim_lock	Write-protect all *otp_trim_registers* in all register maps	0x04 to 0x3F	None
debug_lock	Write-protect the “debug” read/write registers in all register maps	0xC0 to 0xD4	None
i2c_pmb_addr_lock	Write-protect the I2C and PMBus device address registers	0x40	None

6.2 Passwords

Password-based access protection is enabled by setting the protection mode to PASSWORD. The actual passwords are stored in *regmap_common* and are backed by one-time programmable memory (OTP). The default passwords are zero. The password values cannot be read; TDA38740/740A/725/725A/741 will always return 0xFFFF when these registers are read.

Once a password is programmed to a non-zero value, the user must program the matching password value in the xxxx_try_password register. If the password and try_password values do not match, protection is enabled. The operator can attempt to program the correct value into xxx_try_password up to four times, after which they will be locked out. The password attempt count is cleared when power is cycled.

trim_password covers the registers that are defined in the *otp_trim_registers* section of the Common and Loop register maps. It also provides write protection (not read protection) for several “debug” registers that should never be accessed during normal operation.

Note: The trim password provides write protection only.

user_password provides write protection for all registers that are defined in the *otp_cnfg_registers*, *otp_usr_registers*, *read_only_registers*, and *read_write_registers* sections of the Common and Loop register maps. This includes the "debug" registers (also covered by the trim password) and the PMBus command registers.

Note: The user password covers all register map sections (config, trim, and user) for read protection, but only the user sections for write protection.

Table 9 Password protection

Password protection				
		Protected address offsets		
Password	**R/W**	**Description**	**Common**	**PMBus**
trim_password	Read	Not used	–	–
	Write	Covers all trim registers plus the "debug" registers	0x04 to 0x3F 0xC0 to 0xD4	None
user_password	Read	Covers all config, user, and trim registers in all regmaps	0x00 to 0x03 0x04 to 0x3F 0x40 to 0xFF	All
	Write	Covers all config and user registers in all regmaps	0x00 to 0x03 0x40 to 0xFF	All

6.3 Protection modes and sections

The section and mode settings work in combination to define protection for different areas of the register maps. There are separate mode and section fields for read protection and write protection, but their behavior is similar.

A protection mode determines how protection is enabled and disabled. The read_protect_mode and write_protect_mode registers can be set to ALWAYS or PASSWORD. If the mode is PASSWORD, the access protection is enabled if the value in *user_password* does not match the value in *user_try_password*. If they do match, access protection is disabled. If the mode is ALWAYS, protection is enabled irrespective of the passwords.

A protection section specifies one or more regions of the register maps (config, trim, and user).

The section and mode values must both be "active" for protection to be enabled. The section can override the mode, and vice versa. Consider the following examples.

- Section = NONE, mode = ALWAYS: Protection is off for all registers.
- Section = ALL, mode = PASSWORD, user_password is correct: Protection is on for all registers.

Table 10 Protection modes and sections

Protection modes and sections		
		read_protect_mode / write_protect_mode
Value	**Mode**	**Description**
1	ALWAYS	Access protection is enabled; the user password is ignored
0	PASSWORD	Access protection is enabled unless the user password is correct
		write_protect_section
Value	**Section**	**Protected registers**
3	ALL	All config and user registers, including read/write registers that are not backed by OTP
2	VOUT_COMMAND	All registers except VOUT_COMMAND
1	Configuration	All config and user registers that are backed by OTP *(otp_cnfg, otp_usr, and PMBus registers)*
0	NONE	Write protection is disabled
		read_protect_section
Value	**Section**	**Protected registers**
3	ALL	All config, trim, and user registers
2	Telemetry	All except telemetry registers: fail codes, PMBus: VOUT_MODE, STATUS, and telemetry
1	Configuration	All config, trim, and user registers backed by OTP *(otp_cnfg, otp_trim, otp_usr, and PMBus registers)*
0	NONE	Read protection is disabled

6.4 Backdoor keys

It is possible for the protection mode and section to be set such that the user cannot access the device (for example, mode = ALWAYS, section = ALL), including to change the protection mode. Backdoor keys are provided for each section (trim and user) so that the protection can be overridden. Keys are enabled by providing the correct password and by entering test mode (test_in_progress). Thus, if the trim password is correct and *test_in_progress* is active, write, and read protection for the trim section is disabled.

Note: The backdoor keys only override the mode and section settings; they do not override the lock bits.

6.5 write_protect_override

This bit overrides the LOCK_ALWAYS setting in the *write_protect_mode* register. It must be set before setting the protection mode, lets the user be unable to access it. This bit is useful when preparing to program an NVM image with LOCK_ALWAYS mode. It allows the user to set the mode but maintain register access. When this bit is cleared and the image is restored from NVM, then the protection will be in force.

Revision history

Document revision	Date	Description of changes
V 1.0	2024-09-25	Initial release

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